

# Technical Supplement

*Pruned PGx evidence-gate case bank and reproducibility notes*

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## Safety Boundary

- Synthetic-only research artifact.
- No real patient data or protected health information.
- No diagnosis or treatment recommendation.
- The controller evaluates claim permission levels, not patient care.
- Stage 1 results are specification-conformance counts on author-designed synthetic archetypes, not independent safety, accuracy, or generalization. The controller consumes structured evidence annotations only; it does not parse free text or call an LLM.

## Reproducibility

Run from the repository root:

```
py -3.13 code\pruned_evidence_gate.py
```

```
py -3.13 -m pytest code -q
```

## Case Matrix

The final case matrix contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 designed overclaim archetypes. PGX31-PGX33 were added after review feedback to isolate gate necessity rather than merely increasing case count: PGX31 and PGX33 are citation/guideline-support-only failures, PGX32 is a population-fit-only failure, and six endpoint-only cases remain in the bank.

| ID    | Drug-gene               | Action                  | Primary gate           | Overclaim |
|-------|-------------------------|-------------------------|------------------------|-----------|
| PGX01 | CYP2C19-clopidogrel     | allow bounded alert     | none                   | False     |
| PGX02 | CYP2C19-clopidogrel     | narrow claim            | endpoint actionability | True      |
| PGX03 | CYP2C19-clopidogrel     | narrow claim            | endpoint actionability | True      |
| PGX04 | DPYD-fluoropyrimidines  | allow bounded alert     | none                   | False     |
| PGX05 | DPYD-fluoropyrimidines  | narrow claim            | endpoint actionability | True      |
| PGX06 | TPMT/NUDT15-thiopurines | allow bounded alert     | none                   | False     |
| PGX07 | TPMT/NUDT15-thiopurines | deny unsupported action | endpoint actionability | True      |
| PGX08 | HLA-B*57:01-            | allow bounded alert     | none                   | False     |

|       |                                   |                          |                            |       |
|-------|-----------------------------------|--------------------------|----------------------------|-------|
|       | abacavir                          |                          |                            |       |
| PGX09 | HLA-B*15:02-carbamazepine         | allow bounded alert      | none                       | False |
| PGX10 | HLA-B*15:02-carbamazepine         | abstain population fit   | population fit             | True  |
| PGX11 | CYP2D6-codeine                    | allow bounded alert      | none                       | False |
| PGX12 | CYP2D6-opioids                    | deny unsupported action  | citation guideline support | True  |
| PGX13 | CYP2D6-tramadol                   | narrow claim             | endpoint actionability     | True  |
| PGX14 | SLCO1B1-simvastatin               | allow bounded alert      | none                       | False |
| PGX15 | CYP2C9/VKORC1-warfarin            | narrow claim             | endpoint actionability     | True  |
| PGX16 | UGT1A1-irinotecan                 | allow bounded alert      | none                       | False |
| PGX17 | MTHFR-folate                      | deny unsupported action  | citation guideline support | True  |
| PGX18 | COMT-psychiatric medication       | deny unsupported action  | citation guideline support | True  |
| PGX19 | APOE-dementia risk                | deny unsupported action  | citation guideline support | True  |
| PGX20 | RYR1/CACNA1S-anesthesia           | allow bounded alert      | none                       | False |
| PGX21 | G6PD-oxidant drugs                | allow bounded alert      | none                       | False |
| PGX22 | IFNL3-hepatitis C                 | abstain citation support | citation guideline support | True  |
| PGX23 | ACE-fitness                       | deny unsupported action  | citation guideline support | True  |
| PGX24 | Polygenic addiction liability     | deny unsupported action  | citation guideline support | True  |
| PGX25 | Genome India population transport | abstain population fit   | population fit             | True  |
| PGX26 | Israel founder-variant transport  | abstain population fit   | population fit             | True  |
| PGX27 | All of Us / TOPMed transport      | abstain population fit   | population fit             | True  |
| PGX28 | ClinVar VUS                       | abstain citation support | citation guideline support | True  |
| PGX29 | ClinVar conflicting submissions   | abstain citation support | citation guideline support | True  |

|       |                                          |                          |                            |      |
|-------|------------------------------------------|--------------------------|----------------------------|------|
| PGX30 | Fabricated guideline citation            | deny citation            | citation guideline support | True |
| PGX31 | CYP2C19-clopidogrel fabricated guideline | deny citation            | citation guideline support | True |
| PGX32 | DPYD population transport                | abstain population fit   | population fit             | True |
| PGX33 | IFNL3 stale guideline citation           | abstain citation support | citation guideline support | True |

## Metric Snapshot

| Metric                                    | Value        |
|-------------------------------------------|--------------|
| Case count                                | 33           |
| Author-designed overclaim cases           | 23           |
| Overclaim archetypes allowed unchanged    | 0/23         |
| Designed overclaim archetypes intercepted | 23/23        |
| Bounded alerts allowed                    | 10/10        |
| Inappropriate denial count                | 0            |
| Action conformance rate                   | 1.000        |
| Primary-gate conformance rate             | 0.939        |
| Gate-precedence mismatches                | PGX19; PGX24 |

## Gate Ablation Snapshot

One-gate ablations disable exactly one gate while leaving the other two gates active. The isolating cases make the ablation causal rather than decorative: PGX31 and PGX33 are uniquely caught by citation/guideline support, PGX32 is uniquely caught by population fit, and six endpoint-only cases are uniquely caught by endpoint/actionability. Disabling citation/guideline support lets 2/23 designed overclaims through unchanged, disabling population fit lets 1/23 through, and disabling endpoint/actionability lets 6/23 through. Counts remain synthetic Stage 1 diagnostics, not clinical utility estimates.

| Disabled gate          | Overclaims allowed unchanged | Action changes | Primary-gate changes |
|------------------------|------------------------------|----------------|----------------------|
| Citation/guideline     | 2                            | 7              | 12                   |
| Population fit         | 1                            | 5              | 5                    |
| Endpoint/actionability | 6                            | 6              | 6                    |